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AQA A Level Further Mathematics (7367)

Topic **D: Further Algebra and Functions**

Sub topic **D3: Sums of Integers, Squares, and Cubes**

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic	**D: Further Algebra and Functions**
Sub topic	**D3: Sums of Integers, Squares, and Cubes**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ____________________

Date : ____________________

Time : ____________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

(a) Prove by induction that, for all integers $n \geq 1$,

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

(4)

(b) **Hence** show that

$$\sum_{r=1}^{2n} r(r-1)(r+1) = n(n+1)(2n-1)(2n+1)$$

(4)
(Total 8 marks)

Q2.

(a) Prove that

$$6 + 3\sum_{r=1}^{n}(r+1)(r+2) = (n+1)(n+2)(n+3)$$

(6)

(b) Alex substituted a few values of n into the expression $(n + 1)(n + 2)(n + 3)$ and made the statement:

"For all positive integers n,

$$6 + 3\sum_{r=1}^{n}(r+1)(r+2)$$

is divisible by 12."

Disprove Alex's statement.

(2)
(Total 8 marks)

Q3.

(a) Show that

$$\sum_{r=1}^{n} 2r(2r^2 - 3r - 1) = n(n + p)(n + q)^2$$

where p and q are integers to be found.

(6)

(b) Hence find the value of

$$\sum_{r=11}^{20} 2r(2r^2 - 3r - 1)$$

(2)
(Total 8 marks)

Q4.

(a) Show that the equation $4x^3 - x - 540\,000 = 0$ has a root, α, in the interval $51 < \alpha < 52$.

(2)

(b) It is given that $S_n = \sum_{r=1}^{n} (2r - 1)^2$.

(i) Use the formulae for $\sum_{r=1}^{n} r^2$ and $\sum_{r=1}^{n} r$ to show that $S_n = \frac{n}{3}(kn^2 - 1)$, where k is an integer to be found.

(5)

(ii) Hence show that $6S_n$ can be written as the product of three consecutive integers.

(2)

(c) Find the smallest value of N for which the sum of the squares of the first N odd numbers is greater than 180 000.

(2)
(Total 11 marks)

Q5.

(a) Use the formulae for $\sum_{r=1}^{n} r^2$ and $\sum_{r=1}^{n} r^3$ to show that

$$\sum_{r=1}^{n} r^2 (4r - 3) = kn(n + 1)(2n^2 - 1)$$

where k is a constant.

(5)

(b) Hence evaluate

$$\sum_{r=20}^{40} r^2 (4r - 3)$$

(2)
(Total 7 marks)

Q6.

(a) Given that $S_n = \sum_{r=1}^{n} r(3r+1)$ use the formulae for $\sum_{r=1}^{n} r^2$ and $\sum_{r=1}^{n} r$ to show that

$$S_n = n(n + 1)^2$$

(5)

(b) The lowest integer n for which $S_n > 100\,000$ is denoted by N.

Show that

$$N^3 + 2N^2 + N - 100\,000 > 0$$

(1)
(Total 6 marks)

Q7.

(a) Show that

$$\sum_{r=1}^{n} r^3 + \sum_{r=1}^{n} r$$

can be expressed in the form

$$kn(n + 1)(an^2 + bn + c)$$

where k is a rational number and a, b and c are integers.

(4)

(b) Show that there is exactly one positive integer n for which

$$\sum_{r=1}^{n} r^3 + \sum_{r=1}^{n} r = 8 \sum_{r=1}^{n} r^2$$

(5)
(Total 9 marks)

Q8.

The sum to r terms, S_r, of a series is given by

$$S_r = r^2(r + 1)(r + 2)$$

Given that u_r is the rth term of the series whose sum is S_r, show that:

(a) (i) $u_1 = 6$;

(1)

(ii) $u_2 = 42$;

(1)

(iii) $u_n = n(n + 1)(4n - 1)$.

(3)

(b) Show that

$$\sum_{r=n+1}^{2n} u_r = 3n^2(n+1)(5n+2)$$

(3)
(Total 8 marks)

Q9.

It is given that

$$S_n = \sum_{r=1}^{n}(3r^2 - 3r + 1)$$

(a) Use the formulae for $\sum_{r=1}^{n} r^2$ and $\sum_{r=1}^{n} r$ to show that $S_n = n^3$.

(5)

(b) Hence show that $\sum_{r=n+1}^{2n}(3r^2 - 3r + 1) = kn^3$ for some integer k.

(2)
(Total 7 marks)

Q10.

(a) Find

$$\sum_{r=1}^{n}(r^3 - 6r)$$

expressing your answer in the form

$$kn(n+1)(n+p)(n+q)$$

where k is a fraction and p and q are integers.

(5)

(b) It is given that

$$S = \sum_{r=1}^{1000}(r^3 - 6r)$$

Without calculating the value of S, show that S is a multiple of 2008.

(2)
(Total 7 marks)

Q11.

(a) (i) Expand $(2r - 1)^2$.

(1)

(ii) Hence show that

$$\sum_{r=1}^{n}(2r-1)^2 = \tfrac{1}{3}n(4n^2-1)$$

(5)

(b) Hence find the sum of the squares of the odd numbers between 100 and 200.

(4)
(Total 10 marks)

Q12.

Show that

$$\sum_{r=1}^{n}(r^2-r) = kn(n+1)(n-1)$$

where k is a rational number.

(Total 4 marks)

Q13.

(a) Use the formulae

$$\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$$

and $$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

to show that

$$\sum_{r=1}^{n} r^2(r-1) = \frac{1}{12}n(n^2-1)(3n+2)$$

(4)

(b) Use the result from part (a) to find the value of

$$\sum_{r=4}^{11} r^2(r-1)$$

(3)
(Total 7 marks)